

MTHE 351 – Fall 2026

Homework 1

Due Date: Monday, September 21, 2026

1. An urn contains four balls, labeled 1 to 4. Balls are drawn at random one by one, without replacement, until the sum of the numbers on the balls drawn exceeds 4. The sequence of balls drawn is noted. [Note: “exceeds 4” means “strictly larger than 4”.]
 - (a) Write down the sample space for this experiment.
 - (b) Let E be the event “one of the balls drawn is 1”, and F the event “the final sum of numbers on the balls drawn is even”. Give the set of outcomes corresponding to each of the following events
 - (i) “both E and F occur”
 - (ii) “neither E nor F occurs”
 - (iii) “exactly one of the events E, F occurs”
2. Let E, F , and G be three events. Express the following events in using set-theoretic operations:
 - (a) at least two of the three events occur;
 - (b) at most two of the three events occur;
 - (c) exactly two of the three events occur.
3. Which of the following is always true for arbitrary events A, B and C defined on some sample space?
 - (a) $P(A) + P(B) \geq P(A)P(B)$
 - (b) $P(A \cap B) \geq P(A) - P(B^c)$
 - (c) If $P(A) + P(B) = P(C^c)$, then the events A, B and C are mutually exclusive.

You must justify each of your answers with a proof or counterexample, as the case may be. Note that Venn diagrams are unacceptable as proofs, but can be used for counterexamples.

4. Two fair dice both have two of their sides painted red, two painted black, one painted yellow, and one painted white. If we roll this pair of dice, what is the probability that
 - (b) they both land on the same number?
 - (c) they both land on the same color?

5. Let P be a probability defined on a sample space S . For events A of S define $Q(A) = [P(A)]^3$ and $R(A) = P(A)/3$. Is Q a probability on S (i.e., satisfying all probability axioms)? Is R a probability on S ? Justify your answers.
6. Fifty percent of the students at a high school wear neither a ring or a necklace, 40 percent wear a ring, and 30 percent wear a necklace. If one of the students is chosen at random, what is the probability that the student is wearing
- (a) a ring *but not* a necklace?
 - (b) a necklace *but not* a ring?
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